

Chords and diameters

The perpendicular from the centre bisects the chord — and that one fact solves them all.

LESSON 58

1 × 40 MIN

GEOMETRY 8, TEMA 59

STAGE 9 · 7.1

LESSON CLOCK

5'

14'

12'

5'

4'

Fold a paper circle along a chord	5 min
The perpendicular property	14 min
Calculations	12 min
Comparing chords	5 min
Homework	4 min

LEARNING OBJECTIVES

- Use the fact that the perpendicular from the centre bisects a chord.
- Find a chord length from the radius and the distance from the centre.
- Compare chords by their distance from the centre.
- Find the radius from a chord and its distance.

TEXTBOOK

National	Tema 59, pp. 142–143
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

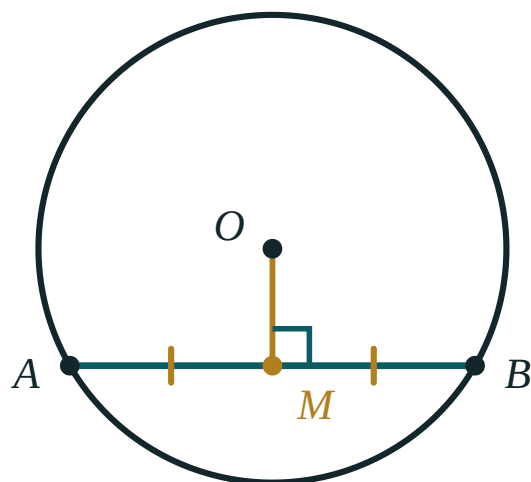
01

Explanation

The perpendicular from the centre

CHORD PROPERTY

The perpendicular drawn from the centre to a chord **bisects** it — and conversely, the line from the centre to the midpoint of a chord is perpendicular to it.



$$OM \perp AB \implies AM = MB$$

The perpendicular from the centre bisects the chord, creating a right triangle with the radius as hypotenuse.

Proof: join O to both ends of the chord. The triangle OAB is isosceles, because $OA = OB = r$. In an isosceles triangle the height from the apex is also the median — so it bisects AB .

THE RIGHT TRIANGLE IT CREATES

$$r^2 = d^2 + \left(\frac{a}{2}\right)^2$$

where a is the chord and d its distance from the centre. Any two of the three give the third.

Comparing chords

Rearranging that relation as $\frac{a}{2} = \sqrt{r^2 - d^2}$ shows the whole story: as d grows, the chord shrinks.

D	CHORD
0	the diameter $2r$ — the longest chord
small	a long chord
$d = r$	the chord shrinks to a point — the line is a tangent

Two consequences: **equal chords are equidistant from the centre**, and of two unequal chords the longer is the nearer. Both follow directly from the formula, with no

new proof.

02 Worked examples

EXAMPLE 1 A chord of length 16 cm is drawn in a circle of radius 10 cm . Find its distance from the centre.

$$\textcircled{1} \quad \frac{a}{2} = 8$$

The perpendicular bisects the chord.

$$\textcircled{2} \quad 10^2 = d^2 + 8^2$$

Pythagoras.

$$\textcircled{3} \quad d^2 = 100 - 64 = 36$$

$$\textcircled{4} \quad d = 6\text{ cm}$$

ANSWER

6 cm

EXAMPLE 2 A chord is 5 cm from the centre of a circle of radius 13 cm . Find its length.

$$\textcircled{1} \quad \left(\frac{a}{2}\right)^2 = 169 - 25 = 144$$

$$\textcircled{2} \quad \frac{a}{2} = 12$$

$$\textcircled{3} \quad a = 24\text{ cm}$$

Do not forget to double.

ANSWER

24 cm

EXAMPLE 3 A chord of 24 cm lies 9 cm from the centre. Find the radius and the diameter.

① $\frac{a}{2} = 12$

② $r^2 = 81 + 144 = 225$

③ $r = 15 \text{ cm}$

④ $d = 30 \text{ cm}$

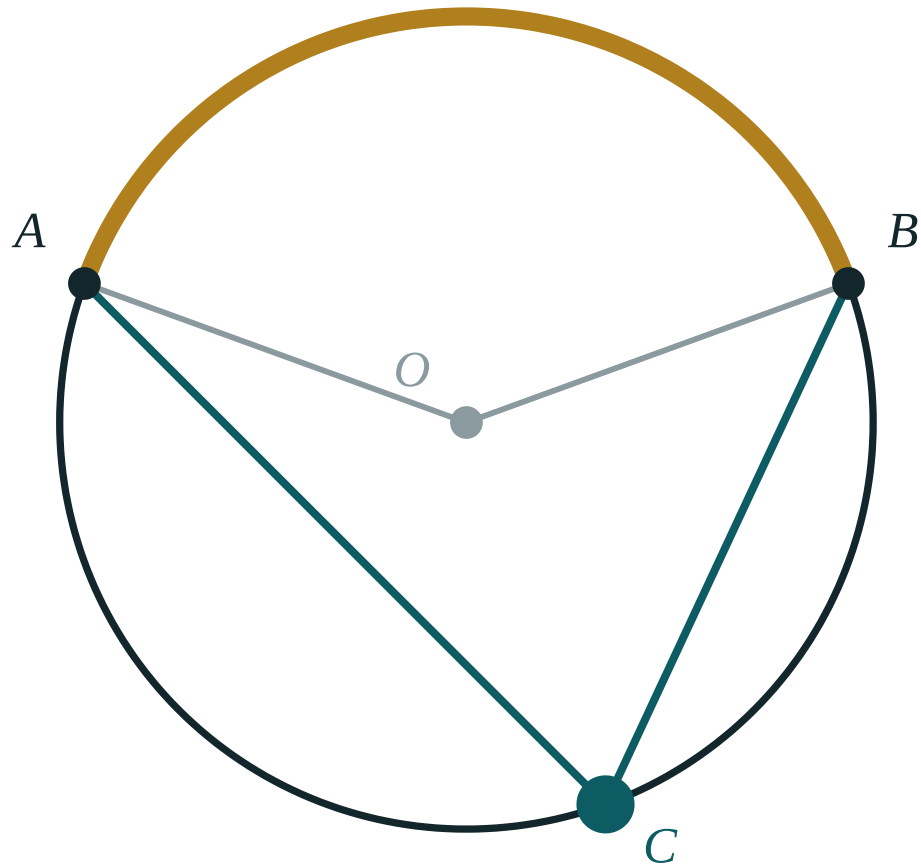
ANSWER

$r = 15 \text{ cm}, d = 30 \text{ cm}$

03 Interactive model

Move the chord and watch the right triangle formed by the radius, the distance and the half-chord.

Chords in a circle

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Chord	Vatar	Хорда
Diameter	Diametr	Диаметр
Perpendicular from the centre	Markazdan perpendikulyar	Перпендикуляр из центра
Bisects	Teng ikkiga bo'ladi	Делит пополам
Distance from the centre	Markazdan masofa	Расстояние от центра
Equal chords	Teng vatarlar	Равные хорды
Half-chord	Yarim vatar	Половина хорды
Longest chord	Eng uzun vatar	Наибольшая хорда

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The perpendicular from the centre to a chord:

is equal to it

bisects it

is a tangent

is a diameter

$r = 5$, chord 8. Its distance from the centre is:

3

4

6

2

$r = 13$, $d = 12$. The chord is:

5

10

25

24

The longest chord of a circle is:

a tangent

a radius

the diameter

any chord

Of two chords, the longer is:

further from the centre

nearer the centre

the same distance

always a diameter

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$, chord 6. Find d .

ANSWER 4

2. $r = 10$, $d = 8$. Find the chord.

ANSWER 12

3. $r = 13$, chord 24. Find d .

ANSWER 5

4. Chord 10, $d = 12$. Find r .

ANSWER 13

5. What is the longest chord of a circle of radius 9?

ANSWER 18

6. $r = 17$, $d = 15$. Find the chord.

ANSWER 16

7. $r = 25$, chord 14. Find d .

ANSWER 24

Medium · the standard question

7 PROBLEMS

1. $r = 15$, chord 18. Find d .

ANSWER 12

2. $r = 20$, $d = 16$. Find the chord.

ANSWER 24

3. Chord 30, $d = 8$. Find r and the diameter.

ANSWER $r = 17$, $d = 34$

4. Two chords of a circle of radius 10 are 6 and 8 from the centre. Find both lengths.

ANSWER 16 and 12

5. $r = 26$, chord 48. Find d .

ANSWER 10

6. A chord of 12 is 8 from the centre. Find the radius.

ANSWER 10

7. Two equal chords are 7 from the centre of a circle of radius 25. Find their length.

ANSWER 48

Hard · stretch and reason

7 PROBLEMS

1. Two parallel chords of a circle of radius 13 are 10 and 24 long. Find the distance between them, in both possible cases.

ANSWER $12 + 5 = 17$ (opposite sides) or $12 - 5 = 7$ (same side)

2. A chord subtends a central angle of 60° in a circle of radius 12. Find its length and its distance from the centre.

ANSWER chord 12 (equilateral triangle), $d = 6\sqrt{3} \approx 10.4$

3. A chord of 16 is 6 from the centre. Find the radius and the central angle it subtends.

ANSWER $r = 10$; $\sin(\alpha/2) = 0.8$, so $\alpha \approx 106^\circ$

4. A circle of radius 10 has two perpendicular chords through a point 6 from the centre, of lengths 16 and 12. Verify with the chord formula.

ANSWER Distances 6 and 8; $6^2 + 8^2 = 100 = r^2$ ✓ — the two distances and the radius form a right triangle

5. Show that equal chords are equidistant from the centre.

ANSWER From $\frac{a}{2} = \sqrt{r^2 - d^2}$, equal a forces equal d in the same circle.

6. A chord is 5 from the centre of a circle of area 169π . Find its length.

ANSWER $r = 13$, chord 24

7. A tunnel is a circular arc 8 m wide at the base and 2 m high in the middle. Find the radius of the circle.

ANSWER $r = 5$ m — $r^2 = (r - 2)^2 + 4^2$

07 Homework

Homework — 5 problems

Geometry 8, Tema 59, pp. 142–143. Draw the right triangle every time.

1. $r = 17$, chord 30. Find d .

2. $r = 25, d = 20$. Find the chord.
3. A chord of 40 is 9 from the centre. Find the radius.
4. Two chords of a circle of radius 15 are 9 and 12 from the centre. Find both lengths.
5. A tunnel arch is 12 m wide and 4 m high in the middle. Find the radius of its circle.